SEMINAR REPORT

A seminar Report on

AI Based Deepfake Detection

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960171	388493	Adnan Anwarul Haque Ansari
2	24610960198	388512	Vivek Umesh Gangurde
3	24610960211	388519	Rohit Aanna Karve
4	24610960216	388524	Bhavesh Ashok Madne

FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

CERTIFICATE

Sr. No	Enrollment Number	Name of student
1	24610960171	Adnan Anwarul Haque Ansari
2	24610960198	Vivek Umesh Gangurde
3	24610960211	Rohit Aanna Karve
4	24610960216	Bhavesh Ashok Madne

of **CO5K** Semester of Diploma in **Computer Engineering** of **S.S.V.P.S's B. S. Deore Polytechnic, Dhule** (Inst.Code: 059) has completed the term work satisfactorily in Course **Seminar Presentation and Project Initiation (Course Code: 315003)** for the academic year **2026 to 2027** as prescribed in the curriculum.

Place : Dhule

Date : .....

Signature:

Name : Shraddha S Hajari

Subject Teacher

Signature:

Name : Nishad Patel

Head of the Department

Seal of Institution

TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	I
	List of Figures	II
	List of Tables	III
1	Introduction	1
2	Literature Review	2
3	Information/Chapters	9
4	Advantages and Disadvantages	14
5	Conclusion	15
6	Project Initiation :	16
	6.1) Description of problem statement.	16
	6.2) Scope and objectives.	16
	6.3) Stake holder	17
	6.4) Platform/ Equipment/ Resources identification.	18
7	Seminar Presentation	19
	Bibliography	29
	References	30

ABSTRACT

Deepfakes are AI-generated or manipulated images, videos, and audio that can appear genuine to viewers. Common methods alter a face, expression, or voice, or generate content that does not correspond to an authentic recording. Synthetic media can support film production, education, accessibility, and creative work; however, malicious use can enable impersonation, fraud, and misinformation. As generated media becomes more realistic and easier to distribute, visual inspection alone may not be sufficient to judge authenticity. AI-based detection therefore plays an important role in digital media verification and digital forensics.

This seminar report reviews research on Convolutional Neural Networks (CNNs), benchmark datasets, hybrid CNN-Transformer models, and graph-based detection approaches. It discusses a general workflow that preprocesses media, extracts facial features, classifies frames, and combines predictions into a video-level estimate. Selected studies are compared in terms of their approach, datasets, objectives, and limitations. The reviewed literature highlights important challenges, including compression, dataset bias, the rapid evolution of generation methods, and poor performance on manipulation types not seen during training. The project outline identifies its problem statement, objectives, stakeholders, and potential software and hardware resources. The proposed system is intended as a screening aid; a model prediction should be checked alongside reliable sources and other forensic evidence rather than treated as a final judgement.

LIST OF FIGURES

Fig. No.	Title	Page No.
3.1	Working of AI-Based Deepfake Detection	10
3.2	Architecture of AI-Based Deepfake Detection	12

LIST OF TABLES

Table No.	Title	Page No.
2.1	Literature Review	6
6.1	Resources	18

CHAPTER 1

INTRODUCTION

Artificial Intelligence (AI) has made it possible to generate and manipulate images, videos, and audio with a high level of realism. Deepfakes are synthetic media in which a face, facial expression, voice, or movement is altered or generated by machine-learning models. Common techniques include face swapping, facial reenactment, face editing, and face synthesis. Such methods can be useful in entertainment and digital production, but the same tools can also be misused.

The spread of realistic fake media creates challenges for public communication, journalism, cybersecurity, and digital forensics. A fabricated clip may be shared widely before its authenticity is checked, while altered media can harm personal reputation or be used in impersonation and fraud. As a result, there is a need for reliable methods that help distinguish genuine recordings from manipulated content.

AI-based deepfake detection systems analyse visual or audio patterns that may differ between genuine and generated media. Image-based approaches commonly use CNNs to identify local texture and blending inconsistencies. More recent studies combine CNN feature extractors with Vision Transformers or graph-based models to learn relationships among facial regions and improve robustness.

A typical detection workflow takes an image or video as input, extracts frames where necessary, detects and crops facial regions, and passes the processed samples to a trained model. The model produces a real/fake score for each sample, and a video-level decision may be obtained by combining frame predictions. Public datasets such as FaceForensics++ and DFDC support model training and evaluation.

The objective of this seminar is to study current deepfake detection techniques, compare selected research papers, identify their advantages and limitations, and outline a practical system workflow. Particular attention is given to generalisation, explainability, and testing under compression or unseen manipulation methods. The final aim is to understand how detection can support, but not replace, responsible media verification.

CHAPTER 2

LITERATURE REVIEW

1. FaceForensics++: Learning to Detect Manipulated Facial Images (2019) - Rössler et al

Introduce a public dataset and benchmark containing genuine videos and manipulated versions produced with DeepFakes, Face2Face, FaceSwap, and NeuralTextures. They evaluate CNN-based detection, including XceptionNet, and show why a shared benchmark is useful. Their results also indicate that compression can hide manipulation traces and make detection more difficult.

2. DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection (2020) - Tolosana et al

Organise facial manipulation into face synthesis, identity swapping, facial-attribute manipulation, and expression swapping. The paper reviews techniques, public datasets, detection results, and open research problems. It is useful for understanding the range of face forgeries, although results from different studies are not always directly comparable because they use different datasets and evaluation settings.

3. MesoNet: a Compact Facial Video Forgery Detection Network (2018) - Afchar et al

Propose two lightweight convolutional networks, Meso-4 and MesoInception-4, for detecting video face tampering. The models learn intermediate-level visual features and combine frame predictions to produce a video-level result. The study reports strong results on the evaluated DeepFake and Face2Face examples, while noting that a detector trained on early manipulation methods may not generalise to newer fakes.

4. Deep Learning for Deepfakes Creation and Detection: A Survey (2019; revised 2021) - Nguyen et al

Rreview how deep learning can generate deepfakes and how manipulated visual media may be detected. They discuss forms such as face swapping, lip-sync, and puppet-master manipulation, alongside generation models and forensic detection techniques. The survey presents challenges and future research directions rather than a single implemented detector, and its overview reflects the methods documented in the literature it covers.

5. A Review of Deep Learning-based Approaches for Deepfake Content Detection (2019; revised 2021) - Passos et al

Systematically review deep-learning approaches for identifying manipulated images and videos. They compare method categories, datasets, evaluation practices, reported strengths, and weaknesses, then identify unresolved issues and future directions. The review highlights the difficulty of comparing results across studies and the need for models that generalise beyond the dataset and manipulation types used during training.

6. A Survey of Deep Fake Detection for Trial Courts (2022) - Celebi, **Liu**, and **Karatoprak**

Examine deepfake detection methods and available datasets with attention to digital forensics and court use. The paper discusses the risks of manipulated images and videos being treated as genuine evidence and reviews approaches to identifying forged media. It underlines that technical detection must be considered alongside provenance, validation, and the wider evidential context.

7. Hybrid Transformer Network for Deepfake Detection (2022) - Khan and **Dang-Nguyen**

Combine feature representations from XceptionNet and EfficientNet-B4 with a Vision Transformer. The hybrid model learns from both CNN feature extractors and uses image augmentations, including face cut-outs, to improve performance and reduce overfitting. It is evaluated on FaceForensics++ and DFDC and reports competitive results, but benchmark performance does not guarantee success on every unseen real-world manipulation.

8. **Self-Supervised Graph Transformer for Deepfake Detection (2023) - Khormali** and **Yuan**

   Propose a detector that uses self-supervised contrastive pre-training, a Vision Transformer feature extractor, graph convolution, and a Transformer discriminator. Facial regions are represented as a graph so the system can learn relationships among parts of the face. The framework also produces relevancy maps to help explain predictions and is tested for cross-dataset, cross-manipulation, and corruption robustness.

9. **Recent Advancements in the Field of Deepfake Detection (2023) - Krueger**, **Vanamala**, and **Dave**

   Survey developments in neural-network-based and other deepfake detection methods. Their review includes audio-focused techniques, visual detectors, and studies on the influence of datasets. The paper stresses that a comprehensive universal detector remains difficult to achieve because deepfake types and data conditions vary, and that training and evaluation data have a substantial effect on reported performance.

10. **FakeFormer: Efficient Vulnerability-Driven Transformers for Generalisable Deepfake Detection (2024) - Nguyen et al**

    Introduce FakeFormer, a Vision Transformer framework that directs attention towards image patches likely to contain blending or manipulation artifacts. The approach is designed to help Transformers capture local inconsistencies that may be overlooked by global attention. The paper evaluates several well-known datasets and reports improved generalisation and computational efficiency, while real-world use still requires testing on new generation techniques.

11. Deepfake Generation and Detection: A Benchmark and Survey (2025) - Pei et al

Provide a broad survey and benchmark of deepfake generation and forgery detection. They cover the transition from variational autoencoders and GANs towards diffusion models, and organise tasks including face swapping, reenactment, talking-face generation, and facial-attribute editing. The paper brings together task definitions, datasets, metrics, and comparisons, while recognising that generation quality and techniques continue to evolve.

12. Exploring Autonomous Methods for Deepfake Detection: A Detailed Survey on Techniques and Evaluation (2026) - Sunil et al

Survey image, video, and audio deepfake detectors published from 2018–2024, reviewing deep-learning, computer-vision, and audio-analysis methods. They note that results depend on datasets and testing conditions.

Table No. 2.1: Literature Review

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Dataset and CNN benchmark	XceptionNet / CNN; face tracking and cropping	FaceForensics++ (about 1,000 videos; 4 manipulation methods)	Provide a public benchmark and compare detectors across several facial-manipulation methods.	Covers older manipulation methods; compression and unseen generators can reduce accuracy.
2	Literature survey and taxonomy	Face-forensics methods; machine learning and deep learning detectors reviewed	Public face-manipulation databases reviewed	Classify manipulation types and compare available datasets and detection approaches.	Survey findings depend on the studies available at publication time; new methods may behave differently.
3	Lightweight deep-learning detector	Meso-4 and MesoInception-4 CNNs; frame-level score aggregation	DeepFake and Face2Face video sets described by the study	Provide a relatively small and efficient detector for facial video forgeries.	Focused on two early manipulation types; visual-only detection may miss newer or audio-based fakes.
4	Survey of generation and detection	Autoencoders, GANs, deep-learning and digital-forensics methods reviewed	Datasets reported throughout the surveyed studies	Summarise deepfake creation techniques, detection methods, challenges, and research trends.	No single detector is proposed; the fast-moving field requires continued review and evaluation.
5	Systematic review of deep-learning detectors	CNNs and other deep-learning approaches reviewed	Public datasets and evaluation setups from reviewed studies	Compare detection categories, report their advantages and drawbacks, and identify research gaps.	Reported results depend on study-specific datasets and metrics; cross-dataset generalisation remains difficult.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
6	Digital-forensics survey	Image forensics, metadata analysis, transfer learning and classifiers reviewed	Deepfake detection datasets reported in the surveyed literature	Review detection methods and datasets relevant to forensic investigation and trial courts.	A model output by itself cannot establish the source, chain of custody, or legal reliability of media.
7	Hybrid CNN-Transformer model	XceptionNet, EfficientNet-B4, Vision Transformer and image augmentation	FaceForensics++ and DFDC	Fuse complementary features and improve detection while reducing overfitting.	Evaluation is benchmark-based; unseen generators, new recording conditions, and compression remain concerns.
8	Self-supervised graph-based detector	Vision Transformer, contrastive learning, graph convolution and Transformer discriminator	Multiple deepfake benchmarks with cross-dataset and corruption tests	Learn more generalisable facial features and improve robustness and interpretability.	The reported results are based on selected experiments; performance on all future manipulations is not assured.
9	Survey of recent detection methods	Neural networks, CNNs, audio-visual methods and alternative approaches reviewed	Datasets used by the studies surveyed, including visual and audio examples	Compare recent strategies and analyse how methods and datasets affect detection.	No universal approach is established; methods and reported results vary across datasets and media types.
10	Vulnerability-driven Vision Transformer	Vision Transformer with learning-based local attention to artifact-prone patches	FaceForensics++, Celeb-DF, WildDeepfake, DFD, DFDCP and DFDC	Improve generalisation and computational efficiency by focusing on local forgery cues.	Depends on visual artifacts and benchmark evaluations; future generators may leave different cues.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
11	Generation/detection survey and benchmark	VAE, GAN, diffusion-based generation and representative detection methods reviewed	Widely used benchmark datasets and metrics	Unify task definitions and evaluate representative generation and detection methods.	Benchmark results cannot cover every real-world context; fast-changing generation methods require updates.
12	Survey of autonomous multimodal detection	CNNs, machine/deep learning, computer vision and audio-analysis methods reviewed	Studies published from 2018–2024 and their evaluation datasets	Review autonomous image, video, and audio detection methods and their evaluation.	Results across modalities and datasets are not directly interchangeable; universal robust detection remains open.

CHAPTER 3
INFORMATION/CHAPTERS

1. Overview of Deepfake Detection:

AI-based deepfake detection uses machine-learning models to examine digital media and estimate whether it has been manipulated. Most facial-video detectors analyse frames containing a face and learn patterns that may indicate synthetic generation or editing. The system output is a likelihood or class label, rather than absolute proof of authenticity.

2. Types of Deepfakes:

Face swapping replaces one person's face with another identity. Facial reenactment transfers expressions or head movements from a source person to a target. Face editing changes selected attributes such as age or expression, while face synthesis generates a face or image that may not correspond to a real person. Audio deepfakes imitate or generate speech and may require specialised audio analysis.

3. Need for Detection:

Realistic fake media can be used to mislead audiences, impersonate people, or circulate false statements. Detection tools can help journalists, researchers, platforms, and investigators prioritise suspicious content for further checking. Since generators continue to improve, detectors should be tested on multiple datasets and not relied upon as the only source of evidence.

4. Working Principle:

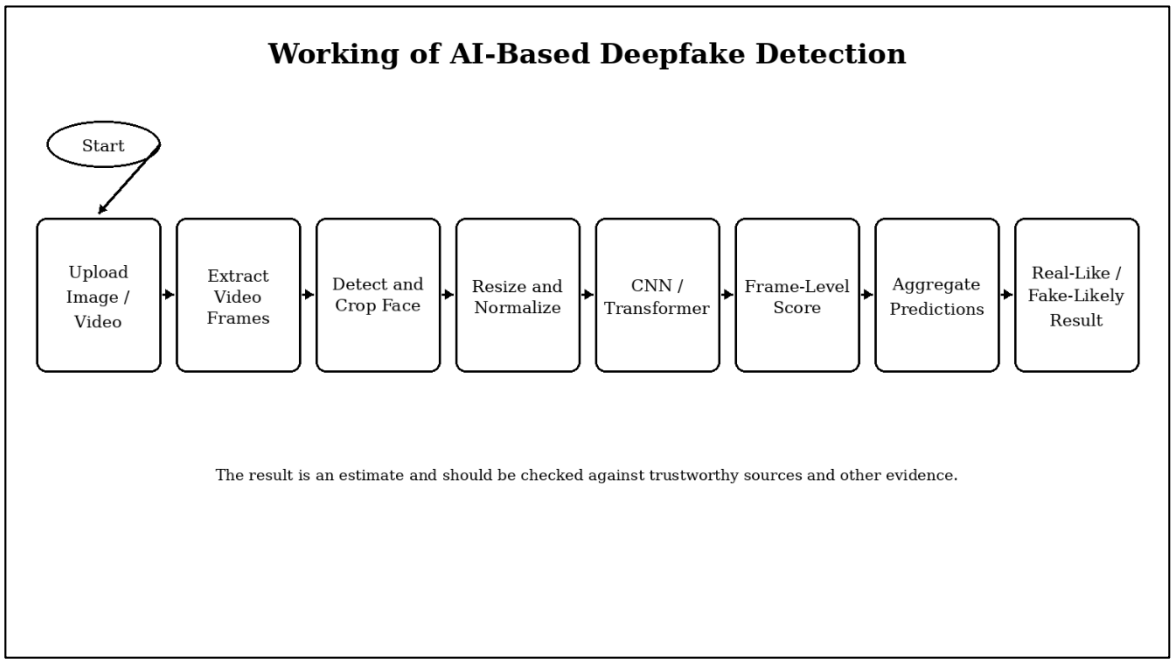


Fig 3.1: Working of AI-Based Deepfake Detection

The general working process of an AI-based deepfake detection system consists of the following steps:

1. **Start:** The process begins when the user selects an image or video for analysis
2. **Input Media:** The system accepts an image or video in a supported file format
3. **Frame Extraction:** For a video, selected frames are sampled so the detector can analyse the clip without processing every frame
4. **Face Detection and Cropping:** The system locates the face and crops the region that will be examined by the model
5. **Pre-processing:** Face samples are resized and normalised to match the model's required input format

6. **Feature Extraction:** The CNN or hybrid model learns visual patterns such as texture, blending boundaries, and regional inconsistencies
7. **Classification:** The detector assigns each analysed frame a score or label indicating whether manipulation is likely
8. **Score Aggregation:** For video, frame-level outputs are combined using a rule such as average score or majority vote
9. **Decision Threshold:** The combined score is compared with a threshold selected during validation
10. **Display Result:** The interface displays the estimated class and, when available, a confidence score
11. **Independent Verification:** Suspicious content is checked against original sources or other forensic evidence
12. **End:** The process finishes after the result and its limitations are shown to the user

5. System Architecture and Main Components:

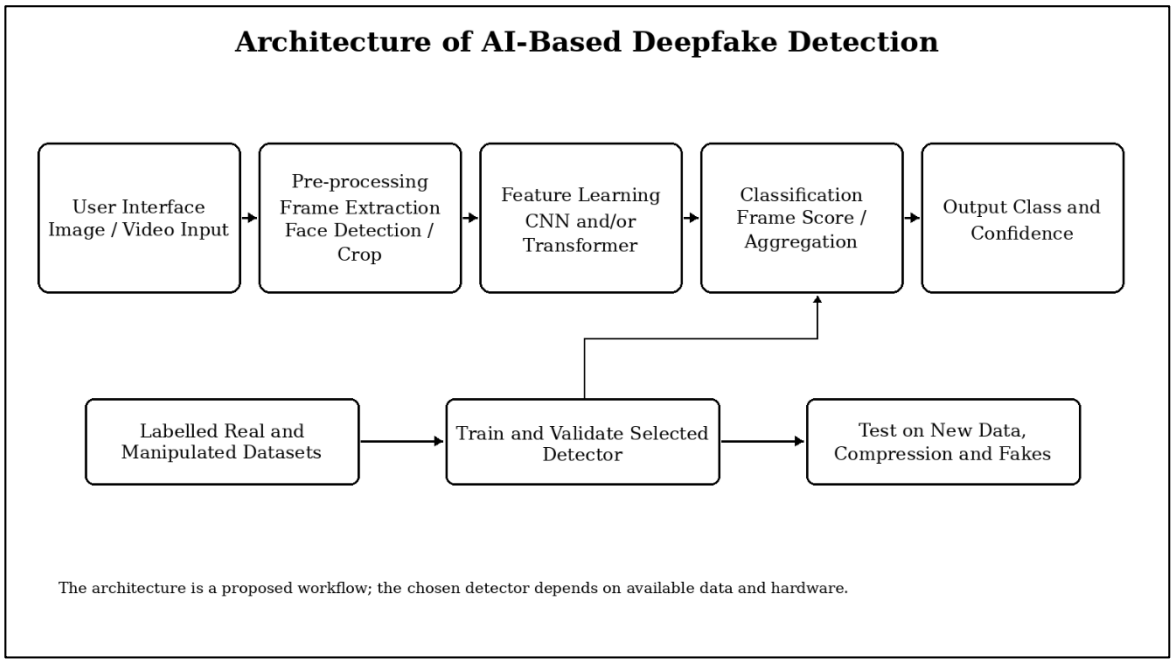


Fig 3.2: Architecture of AI-Based Deepfake Detection

The main components include the User Interface, Media Pre-processing Module, Face Detection and Cropping Module, Deepfake Detection Model, Prediction Aggregation Module, and Results Display Module. The training workflow also requires labelled data, model validation, and testing on videos that differ from the training samples.

6. Technologies and Techniques Used:

Python can be used to develop the detection pipeline. OpenCV supports frame extraction, image resizing, and pre-processing. PyTorch or TensorFlow can run a trained CNN or hybrid model such as XceptionNet with a Transformer. Dataset selection depends on access permissions and project resources; FaceForensics++ and DFDC are examples of research benchmarks. Testing should include compressed and unseen samples where possible.

7. Example of Deepfake Detection:

Consider a short video that appears to show a public speaker making an unusual statement. The user submits the video to the proposed system for screening. The software extracts selected frames, detects the face, prepares cropped samples, and passes them to the trained model.

The model evaluates visual features and produces a score for each frame. If multiple frames receive high manipulated-media scores, the aggregation module may label the clip as "manipulated-likely" and display the confidence estimate. The result helps identify content that needs review, but it does not independently establish who created the clip or whether it was deliberately misleading.

8. Interpreting the Result:

A real-like prediction means that the model did not detect strong signs of manipulation in the analysed samples; it does not guarantee that the media is genuine. A manipulated-likely prediction can also be wrong, especially with poor lighting, low resolution, heavy compression, unusual camera angles, or new generation methods. Important media should be checked using source verification, metadata when available, and independent evidence.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Faster Screening: A trained detector can analyse many frames more quickly than relying only on manual inspection.

2. Consistent Analysis: The same model can apply its learned criteria to each input, supporting repeatable first-stage screening.

3. Useful for Large Media Collections: Automated tools can help prioritise suspicious items for additional review.

4. Adaptable Methods: CNNs, Transformers, and graph-based methods can be evaluated and improved as new research becomes available.

5. Benchmark Evaluation: Public research datasets allow performance to be compared under controlled conditions.

4.2 Disadvantages:

1. False Predictions: The system can classify genuine media as fake or fail to identify manipulated content.

2. Limited Generalisation: A model trained on one dataset may perform poorly on unseen generators or different recording conditions.

3. Compression Effects: Resizing, blur, and video compression can remove useful traces and lower accuracy.

4. Computational Requirements: Training large models may require substantial memory, labelled data, and GPU resources.

5. Evolving Technology: Detection methods must be evaluated and updated as generative techniques change.

CHAPTER 5
CONCLUSION

AI-based deepfake detection is an important area of digital forensics because synthetic media can be difficult to distinguish from genuine recordings. Research has progressed from CNN-based detectors and benchmark datasets to hybrid Transformer and graph-based approaches that aim to learn richer and more generalisable features.

The reviewed papers show that performance depends on the dataset, manipulation method, image quality, and evaluation procedure. Cross-dataset testing, robustness to compression, explainability, and testing on unseen content remain important challenges. No single detector can be assumed to identify every deepfake correctly.

A practical detection workflow can preprocess media, analyse facial regions, combine frame-level predictions, and present a result for review. Such a tool can support verification when its limitations are clearly stated and results are checked against trustworthy sources. Continued research and responsible use are necessary as generation technology evolves.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement:

Recent AI advances make it possible to generate realistic images and videos or alter faces and expressions. Such media can spread misinformation or enable impersonation and fraud, while subtle edits and compression make manual inspection difficult.

Detectors can perform well on familiar datasets but may struggle with new manipulation methods, different recording conditions, or low-quality video. A practical system should analyse supplied media, estimate the likelihood of manipulation, and communicate uncertainty clearly.

The project outlines a workflow for preprocessing images or video, analysing face regions with a trained model, and producing an estimated result. Evaluation will use suitable data and metrics; the result is not a definitive forensic judgement.

6.2 Scope of the Project:

The project scope is a study or small prototype that accepts images or videos, samples frames, prepares face crops, runs a trained detector, and displays an estimated label. A CNN baseline may be compared with a more advanced model where resources permit. Large-scale deployment and guaranteed detection of all deepfakes are outside the scope.

Objectives of the Project:

1. Study deepfake types and their potential misuse.
2. Review CNN, Transformer, and graph-based detection approaches.
3. Outline media input, preprocessing, detection, and result reporting.
4. Evaluate performance using suitable metrics and labelled data.
5. Identify issues caused by compression, unseen fakes, and dataset bias.
6. Emphasize that model scores support, but do not replace, verification.

6.3 Stakeholders:

The stakeholders are the people and groups who develop, evaluate, use, or are affected by an AI-based deepfake detection system.

1. **End Users:** Individuals or organisations that want to screen suspicious images and videos before sharing or relying on them

2. **Developers:** The team responsible for designing the interface, preprocessing pipeline, model integration, and test procedures

3. **Researchers:** People who study media forensics, deep learning models, datasets, and evaluation methods

4. **Media and Fact-Checking Teams:** Journalists or verification staff who may use a model result as one signal for further investigation

5. **Platform and Security Teams:** Groups that may need to review suspicious media while considering privacy and misuse risks

6. **Project Guide and Evaluators:** Faculty members who guide the seminar and evaluate the report, presentation, and proposed system design

6.4 Platform / Equipment / Resources:

1. Platform:

The project can be studied and prototyped on Windows or Linux using Python. Visual Studio Code can be used for development, and a framework such as PyTorch or TensorFlow can load or train the selected detection model. A local environment is suitable for a basic prototype; larger experiments may require access to a GPU-enabled machine.

2. Equipment:

- **Computer or Laptop:** For coding, testing, and running the detector
- **Processor:** A modern multi-core processor for general development and preprocessing
- **RAM:** At least 8 GB is recommended for a basic workflow; larger models may require more memory
- **GPU (if available):** Helpful for model training and faster inference; it is not mandatory for a small prototype
- **Internet and Storage:** Needed to access permitted datasets, install libraries, and store model files and test media

3. Resources (Table 6.1):

Resources	Name	Purpose
Operating System	Windows 11 / Linux	Host development tools and run the local testing environment.
Programming Language	Python 3.11	Implement preprocessing, model inference, and evaluation scripts.
Code Editor	Visual Studio Code	Code editing, debugging, and project organisation.
Deep Learning Framework	PyTorch or TensorFlow	Load, train, or evaluate supported neural-network models.
Image Processing	OpenCV	Extract video frames, resize images, and prepare face crops.
Detection Model	XceptionNet / selected CNN	Provide a baseline for classifying facial samples.
Research Dataset	FaceForensics++ / DFDC	Provide labelled genuine and manipulated media for evaluation, subject to access terms.
Hardware	Laptop; GPU if available	Run experiments; a GPU may be needed for larger training jobs.

SEMINAR PRESENTATION

	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	**SPI – 315003**	**Date :**

Topic : AI Based Deepfake Detection

Sr. No.	Name of Student
1	Adnan Anwarul Haque Ansari
2	Vivek Umesh Gangurde
3	Rohit Aanna Karve
4	Bhavesh Ashok Madne

Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information/Chapters
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6. 1 Description of problem statement.
   - 6. 2 Scope and objectives.
   - 6. 3 Stake holder
   - 6. 4 Platform/ Equipment/ Resources identification.
   - Seminar Presentation
7. Bibliography
8. References

2

Abstract

Deepfakes are AI-generated or manipulated images, videos, and audio that can appear genuine to viewers, created by altering a face, expression, or voice. While synthetic media can support film production, education, and creative work, malicious use can enable impersonation, fraud, and misinformation. As generated media becomes more realistic, visual inspection alone is often insufficient to judge authenticity, making AI-based detection important for digital media verification and forensics. This seminar reviews research on Convolutional Neural Networks (CNNs), benchmark datasets, hybrid CNN-Transformer models, and graph-based detection approaches, and discusses a workflow that preprocesses media, extracts facial features, classifies frames, and aggregates predictions into a video-level estimate. Key challenges include compression, dataset bias, and poor performance on unseen manipulation types. The proposed system is intended as a screening aid, to be checked alongside reliable sources rather than treated as a final judgement.

1. Introduction

- AI has made it possible to generate and manipulate images, videos, and audio with high realism.
- Deepfakes alter or generate a face, expression, voice, or movement using machine-learning models.
- Common techniques include face swapping, facial reenactment, face editing, and face synthesis.
- Realistic fake media creates challenges for journalism, cybersecurity, and digital forensics.
- AI-based detection analyses visual or audio patterns to separate genuine media from manipulated content.
- This seminar studies detection techniques, compares research papers, and outlines a practical system workflow.

2. Literature Review

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Dataset and CNN benchmark	XceptionNet / CNN; face tracking and cropping	FaceForensics++ (about 1,000 videos; 4 manipulation methods)	Provide a public benchmark and compare detectors across several facial-manipulation methods.	Covers older manipulation methods; compression and unseen generators can reduce accuracy.
2	Literature survey and taxonomy	Face-forensics methods; machine learning and deep learning detectors reviewed	Public face-manipulation databases reviewed	Classify manipulation types and compare available datasets and detection approaches.	Survey findings depend on the studies available at publication time; new methods may behave differently.
3	Lightweight deep-learning detector	Meso-4 and MesoInception-4 CNNs; frame-level score aggregation	DeepFake and Face2Face video sets described by the study	Provide a relatively small and efficient detector for facial video forgeries.	Focused on two early manipulation types; visual-only detection may miss newer or audio-based fakes.

5

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	Survey of generation and detection	Autoencoders, GANs, deep-learning and digital-forensics methods reviewed	Datasets reported throughout the surveyed studies	Summarise deepfake creation techniques, detection methods, challenges, and research trends.	No single detector is proposed; the fast-moving field requires continued review and evaluation.
5	Systematic review of deep-learning detectors	CNNs and other deep-learning approaches reviewed	Public datasets and evaluation setups from reviewed studies	Compare detection categories, report their advantages and drawbacks, and identify research gaps.	Reported results depend on study-specific datasets and metrics; cross-dataset generalisation remains difficult.
6	Digital-forensics survey	Image forensics, metadata analysis, transfer learning and classifiers reviewed	Deepfake detection datasets reported in the surveyed literature	Review detection methods and datasets relevant to forensic investigation and trial courts.	A model output by itself cannot establish the source, chain of custody, or legal reliability of media.

6

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Hybrid CNN-Transformer model	XceptionNet, EfficientNet-B4, Vision Transformer and image augmentation	FaceForensics++ and DFDC	Fuse complementary features and improve detection while reducing overfitting.	Evaluation is benchmark-based; unseen generators, new recording conditions, and compression remain concerns.
8	Self-supervised graph-based detector	Vision Transformer, contrastive learning, graph convolution and Transformer discriminator	Multiple deepfake benchmarks with cross-dataset and corruption tests	Learn more generalisable facial features and improve robustness and interpretability.	The reported results are based on selected experiments; performance on all future manipulations is not assured.
9	Survey of recent detection methods	Neural networks, CNNs, audio-visual methods and alternative approaches reviewed	Datasets used by the studies surveyed, including visual and audio examples	Compare recent strategies and analyse how methods and datasets affect detection.	No universal approach is established; methods and reported results vary across datasets and media types.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Vulnerability-driven Vision Transformer	Vision Transformer with learning-based local attention to artifact-prone patches	FaceForensics++, Celeb-DF, WildDeepfake, DFD, DFDCP and DFDC	Improve generalisation and computational efficiency by focusing on local forgery cues.	Depends on visual artifacts and benchmark evaluations; future generators may leave different cues.
11	Generation/detection survey and benchmark	VAE, GAN, diffusion-based generation and representative detection methods reviewed	Widely used benchmark datasets and metrics	Unify task definitions and evaluate representative generation and detection methods.	Benchmark results cannot cover every real-world context; fast-changing generation methods require updates.
12	Survey of autonomous multimodal detection	CNNs, machine/deep learning, computer vision and audio-analysis methods reviewed	Studies published from 2018-2024 and their evaluation datasets	Review autonomous image, video, and audio detection methods and their evaluation.	Results across modalities and datasets are not directly interchangeable; universal robust detection remains open.

3. Information to seminar topic

- **Technology :** Convolutional Neural Networks (CNNs), Vision Transformers, graph-based models
- **Main purpose :** To estimate whether an image or video has been manipulated to produce a deepfake.
- **Key concept :** Learning visual patterns such as texture, blending boundaries, and facial inconsistencies.
- **Approach :** Extracts frames, detects and crops faces, classifies each sample, and aggregates scores into a video-level result.
- **Main benefit :** Faster, consistent first-stage screening to support digital media verification and forensics.
- **Application :** Can be applied to screening suspicious videos, journalism verification, and platform content moderation.

9

4. Advantages / Disadvantages

Advantages

- Faster screening: analyses many frames more quickly than manual inspection.
- Consistent analysis: applies the same learned criteria to every input.
- Useful for large collections: helps prioritise suspicious items for review.
- Adaptable methods: CNNs, Transformers, and graph-based methods can be improved over time.
- Benchmark evaluation: public datasets allow performance to be compared fairly.

Disadvantages

- False predictions: genuine media can be misclassified as fake, or vice versa.
- Limited generalisation: a model trained on one dataset may perform poorly on unseen generators.
- Compression effects: resizing, blur, and compression can remove useful detection traces.
- Computational requirements: training large models needs substantial memory and GPU resources.
- Evolving technology: detection methods must be updated as generation techniques change.

12

5. Conclusion

AI-based deepfake detection is an important area of digital forensics because synthetic media can be difficult to distinguish from genuine recordings. Research has progressed from CNN-based detectors to hybrid Transformer and graph-based approaches that learn richer, more generalisable features. A practical detection workflow can preprocess media, analyse facial regions, combine frame-level predictions, and present a result for review supporting verification without replacing responsible, source-checked judgement.

13

6. Project Initiation

The AI-Based Deepfake Detection project was initiated to address the growing risk of realistic synthetic media, where a face, expression, or voice can be altered or generated using machine-learning models. Such content can mislead audiences, enable impersonation, and spread misinformation.

The project's aim is to study a system that preprocesses media, detects and analyses facial regions with a trained model, and reports an estimated result. The output is intended as a screening aid, to be checked alongside reliable sources rather than treated as a final judgement.

14

6. 1 Description of problem statement.

Recent AI advances make it possible to generate realistic images and videos or alter faces and expressions.

Such media can spread misinformation or enable impersonation and fraud, while compression makes manual inspection difficult.

Detectors can perform well on familiar datasets but may struggle with new manipulation methods or low-quality video.

A practical system should analyse supplied media, estimate the likelihood of manipulation, and communicate uncertainty clearly.

15

6. 2 Scope and objectives.

Objectives

- Study deepfake types and their potential misuse.
- Review CNN, Transformer, and graph-based detection approaches.
- Outline media input, preprocessing, detection, and result reporting.
- Evaluate performance using suitable metrics and labelled data.
- Identify issues caused by compression, unseen fakes, and dataset bias.
- Emphasise that model scores support, but do not replace, verification.

Scope

- Accepts images or videos as input for screening.
- Samples frames and prepares face crops from video input.
- Runs a trained detector such as a CNN baseline.
- Compares a CNN baseline with a more advanced model where resources permit.
- Produces an estimated real/fake label for review.

- Excludes large-scale deployment and guaranteed detection of all deepfakes.

16

6. 3 Stake holder

- **End Users:** Individuals or organisations that want to screen suspicious images and videos before sharing or relying on them.
- **Developers:** The team responsible for designing the interface, preprocessing pipeline, model integration, and test procedures.
- **Researchers:** People who study media forensics, deep learning models, datasets, and evaluation methods.
- **Media and Fact-Checking Teams:** Journalists or verification staff who may use a model result as one signal for further investigation.
- **Platform and Security Teams:** Groups that may need to review suspicious media while considering privacy and misuse risks.
- **Project Guide and Evaluators:** Faculty members who guide the seminar and evaluate the report, presentation, and proposed system design.

17

7. Seminar Presentation

1. **Motivation:** Introduce the topic and explain why realistic synthetic media creates a need for verification.
2. **Deepfake Types:** Define deepfakes and describe face swapping, facial reenactment, face editing, and face synthesis.
3. **Detection Workflow:** Explain the basic CNN-based detection workflow and the role of face preprocessing.
4. **Benchmark Dataset:** Summarise FaceForensics++ as a benchmark for training and comparing detection models.
5. **Hybrid Models:** Describe the hybrid CNN-Transformer approach and why feature fusion may be useful.
6. **Graph-Based Methods:** Discuss graph-based and self-supervised methods that improve robustness and interpretability.
7. **Proposed Workflow:** Present the proposed workflow and architecture diagrams included in this report.

18

6. 4 Platform/ Equipment/ Resources

Hardware Requirements:

- Computer or laptop for coding, testing, and running the detector.
- RAM: at least 8 GB recommended; larger models may need more.
- GPU (if available): helpful for training and faster inference, not mandatory for a small prototype.

Software Requirements:

- Operating system: Windows or Linux.
- Programming language: Python 3.11.
- Deep learning framework: PyTorch or TensorFlow.
- Image processing: OpenCV for frame extraction and face cropping.
- Code editor: Visual Studio Code.

Other Resources:

- Research datasets such as FaceForensics++ and DFDC (subject to access terms).
- Internet and storage to install libraries and store model files and test media.
- Testing on compressed and unseen samples where possible.

19

Bibliography

- **Rossler et al. (2019) - FaceForensics++: Learning to Detect Manipulated Facial Images, ICCV.**
- Tolosana et al. (2020) - DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection.
- Afchar et al. (2018) - MesoNet: a Compact Facial Video Forgery Detection Network, WIFS.
- Nguyen et al. (2021) - Deep Learning for Deepfakes Creation and Detection: A Survey.
- Passos et al. (2024) - A Review of Deep Learning-based Approaches for Deepfake Content Detection.

- **Celebi, Liu and Karatoprak (2022) - A Survey of Deep Fake Detection for Trial Courts.**

20

Bibliography (contd.)

- **Khan and Dang-Nguyen (2022) - Hybrid Transformer Network for Deepfake Detection.**
- Khormali and Yuan (2023) - Self-Supervised Graph Transformer for Deepfake Detection.
- Krueger, Vanamala and Dave (2023) - Recent Advancements in the Field of Deepfake Detection.
- Nguyen et al. (2024) - FakeFormer: Efficient Vulnerability-Driven Transformers for Deepfake Detection.
- Pei et al. (2026) - Deepfake Generation and Detection: A Benchmark and Survey.
- **Sunil et al. (2025) - Exploring Autonomous Methods for Deepfake Detection: A Survey.**

References

- **Research Papers**

1. FaceForensics++: Learning to Detect Manipulated Facial Images
2. DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection
3. MesoNet: a Compact Facial Video Forgery Detection Network
4. Deep Learning for Deepfakes Creation and Detection: A Survey
5. A Review of Deep Learning-based Approaches for Deepfake Content Detection
6. A Survey of Deep Fake Detection for Trial Courts
7. Hybrid Transformer Network for Deepfake Detection
8. Self-Supervised Graph Transformer for Deepfake Detection
9. Recent Advancements in the Field of Deepfake Detection
10. FakeFormer: Efficient Vulnerability-Driven Transformers for Generalisable Deepfake Detection
11. Deepfake Generation and Detection: A Benchmark and Survey
12. Exploring Autonomous Methods for Deepfake Detection: A Detailed Survey on Techniques and Evaluation

BIBLIOGRAPHY

1. Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., and Nießner, M. "FaceForensics++: Learning to Detect Manipulated Facial Images." Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019. doi:10.1109/ICCV.2019.00009.

2. Tolosana, R., Vera-Rodriguez, R., Fierrez, J., Morales, A., and Ortega-Garcia, J. "DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection." Information Fusion, vol. 64, pp. 131–148, 2020. doi:10.1016/j.inffus.2020.06.014.

3. Afchar, D., Nozick, V., Yamagishi, J., and Echizen, I. "MesoNet: a Compact Facial Video Forgery Detection Network." 2018 IEEE International Workshop on Information Forensics and Security (WIFS), 2018. doi:10.1109/WIFS.2018.8630761.

4. Nguyen, T. T., Nguyen, Q. V. H., Nguyen, C. M., Nguyen, D., Nguyen, D. T., and Nahavandi, S. "Deep Learning for Deepfakes Creation and Detection: A Survey." arXiv:1909.11573, revised 2021.

5. Passos, L. A., Jodas, D., Costa, K. A. P., Souza Júnior, L. A., Rodrigues, D., Del Ser, J., Camacho, D., and Papa, J. P. "A Review of Deep Learning-based Approaches for Deepfake Content Detection." arXiv:2202.06095, revised 2024.

6. Celebi, N., Liu, Q., and Karatoprak, M. "A Survey of Deep Fake Detection for Trial Courts." Computer Science & Information Technology, pp. 227–238, 2022. doi:10.5121/csit.2022.120919.

7. Khan, S. A., and Dang-Nguyen, D.-T. "Hybrid Transformer Network for Deepfake Detection." arXiv:2208.05820, 2022.

8. Khormali, A., and Yuan, J.-S. "Self-Supervised Graph Transformer for Deepfake Detection." arXiv:2307.15019, 2023.

9. Krueger, N., Vanamala, M., and Dave, R. "Recent Advancements in the Field of Deepfake Detection." arXiv:2308.05563, 2023.

10. Nguyen, D., Astrid, M., Ghorbel, E., and Aouada, D. "FakeFormer: Efficient Vulnerability-Driven Transformers for Generalisable Deepfake Detection." arXiv:2410.21964, version 2, 2024.

11. Pei, G., Zhang, J., Hu, M., Zhang, Z., Wang, C., Wu, Y., Zhai, G., Yang, J., and Tao, D. "Deepfake Generation and Detection: A Benchmark and Survey." arXiv:2403.17881, revised 2026.

12. Sunil, R., Mer, P., Diwan, A., Mahadeva, R., and Sharma, A. "Exploring Autonomous Methods for Deepfake Detection: A Detailed Survey on Techniques and Evaluation." Heliyon, vol. 11, e42273, 2025. doi:10.1016/j.heliyon.2025.e42273.

REFERENCES

[1] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner, "FaceForensics++: Learning to Detect Manipulated Facial Images," Proc. IEEE/CVF International Conference on Computer Vision (ICCV), 2019, doi:10.1109/ICCV.2019.00009.

[2] R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales, and J. Ortega-Garcia, "DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection," Information Fusion, vol. 64, pp. 131–148, 2020, doi:10.1016/j.inffus.2020.06.014.

[3] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen, "MesoNet: a Compact Facial Video Forgery Detection Network," 2018 IEEE International Workshop on Information Forensics and Security (WIFS), 2018, doi:10.1109/WIFS.2018.8630761.

[4] T. T. Nguyen et al., "Deep Learning for Deepfakes Creation and Detection: A Survey," arXiv:1909.11573, 2021.

[5] L. A. Passos et al., "A Review of Deep Learning-based Approaches for Deepfake Content Detection," arXiv:2202.06095, revised 2024.

[6] N. Celebi, Q. Liu, and M. Karatoprak, "A Survey of Deep Fake Detection for Trial Courts," Computer Science & Information Technology, pp. 227–238, 2022, doi:10.5121/csit.2022.120919.

[7] S. A. Khan and D.-T. Dang-Nguyen, "Hybrid Transformer Network for Deepfake Detection," arXiv:2208.05820, 2022.

[8] A. Khormali and J.-S. Yuan, "Self-Supervised Graph Transformer for Deepfake Detection," arXiv:2307.15019, 2023.

[9] N. Krueger, M. Vanamala, and R. Dave, "Recent Advancements in the Field of Deepfake Detection," arXiv:2308.05563, 2023.

[10] D. Nguyen, M. Astrid, E. Ghorbel, and D. Aouada, "FakeFormer: Efficient Vulnerability-Driven Transformers for Generalisable Deepfake Detection," arXiv:2410.21964, 2024.

[11] G. Pei et al., "Deepfake Generation and Detection: A Benchmark and Survey," arXiv:2403.17881, revised 2026.

[12] R. Sunil, P. Mer, A. Diwan, R. Mahadeva, and A. Sharma, "Exploring Autonomous Methods for Deepfake Detection: A Detailed Survey on Techniques and Evaluation," Heliyon, vol. 11, e42273, 2025, doi:10.1016/j.heliyon.2025.e42273.

SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Name of student
1	24610960171	Adnan Anwarul Haque Ansari
2	24610960198	Vivek Umesh Gangurde
3	24610960211	Rohit Aanna Karve
4	24610960216	Bhavesh Ashok Madne

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388493	388493	388493	388493
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : ----------

(Course Expert/s)